FRACTIONAL OPTIMAL CONTROL PROBLEM MODELING BOVINE TUBERCULOSIS AND RABIES CO-INFECTION

Dr. Boubacar DIALLO
Prof. Jeconia Abonyo OKELO, Nairobi
Dr. Shaibu OSMAN, Accra

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Abstract

Problem

- Rabies virus and bovine tuberculosis continue to pose a significant global health burden;
- The aim of this research is to formulate a fractional order model for the evolotion and co-infection of bovine tuberculosis and rabies that describes the transmission dynamics of the diseases and also show the suitable control strategies that prevent the spread of them;
- Forward-Backward Sweep Method, was used to solve the optimality system
- Culling and vaccination are the two control strategies considered in this research;
- A detailed mathematical analysis was carried out to show that the models are well-posed;
- The effects of the co-infection was discussed
- Numerical simulations were also carried out to validate the models and they were presented graphically and in tabular form.

Outline

Background of the Study

bovine Tuberculosis (bTB)

Bovine tuberculosis (bTB) is a zoonotic disease of global importance endemic in African buffalo (Syncerus caffer) in sub-Saharan Africa [V et al., 2001]

Lack of routine surveillance data

Due to a lack of routine surveillance data, the burden of zoonotic TB is most likely underestimated; however, in 2016 there were an estimated 147,000 new cases of zoonotic TB in people, and 12,500 deaths due to the disease globally [Global tuberculosis report. Geneva: World Health Organization; 2023].

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In Africa, however, bovine[Podlubny, 1999] TB represents a potential health hazard to both animals and humans, as nearly 85% of cattle and 82% of the human population live in areas where the disease is prevalent or only partially controlled[Ayele et al., 2004].

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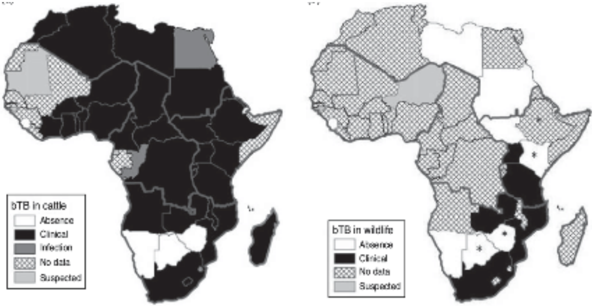

Figure 1: Distribution map of bovine tuberculosis in Africa during 1996-2011. Source: [De Garine-Wichatitsky et al., 2013]

Background of the Study

Rabies

Rabies is one of the zoonoses, these diseases affect animals and can be transmitted to humans via a bite or a scratch.
Rabies is a pathology caused by the lyssavirus, an RNA virus.These viruses are present world-wide but endemic in Africa and Asia.

Symptomes

- In animals, the symptoms depend on the species concerned.
- In humans the disease usually progresses in two phases: **Prodromal phase, and State phase**
- The natural evolution of rabies is toward a coma followed inexorably by death.

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55,000 persons die of rabies

Worldwide, it is estimated that approximately 55,000 persons die of rabies each year according to CDC.

An estimated 21 476 human deaths occur each year in Africa

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The number of reported cases of rabies in Mali is eight (8). Jul 2023.
As for dog bites, the tally is 690 reported cases across the country.
Malian authorities have expressed their determination to eradicate rabies by 2029.

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Hospitalized human rabies victim in restraints

Figure 2: Hospitalized human rabies victim in restraints
Source: Centers for Disease Control and Prevention.

Literature Review: Bovine Tuberculosis

- A deterministic model for studying the transmission dynamics of bovine tuberculosis in a single cattle herd is presented and qualitatively analyzed in [Mathews et al., 2006] and [Agusto et al., 2011]
- In South Africa [Phepha, 2015] to understand the transmission dynamics of M. bovis, mathematical epidemiological models are developed. The findings have very vital implications for bovine tuberculosis control. If bovine tuberculosis is to be eliminated, there is need to develop tests that can detect buffalo carriers from buffalo population.
- To examine the transmission dynamics of bTB and its control methods. [Liu et al., 2016] established a dynamical model for tuberculosis of humans and cows. For the model, they firstly give the basic reproduction number and discuss the dynamical behaviors of the model.

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Literature Review: Bovine Tuberculosis

In [Shirima Sabini et al., 2020], a mathematical deterministic model is formulated to study the transmission dynamics of bTB in humans and animals. The basic reproduction number R_0 is calculated to determine the behavior of the disease. Stability analysis shows that there is a possibility for disease-free equilibrium and endemic equilibrium to co-exist when R0 = 1.

Literature Review : Rabies

- In order to understand the epidemiological characteristics of fox rabies in Europe[Anderson et al., 1981] originally presented a deterministic model with susceptible, exposed, and recovered groups.
- The dynamic shift in raccoon rabies in the US was predicted by . [Childs et al., 2000]. They analyzed the dynamics of local epizootics at the county level by examining a database spanning more than 20 years and including 35,387 rabid raccoons.
- In African cities, Zinsstag et al.[Zinsstag et al., 2009] examined the dynamics of canine and human rabies transmission and their economics. They argued that it was more economical to combine a canine immunization campaign with a human vaccination program.
- To examine the dynamics of rabies transmission in Ghana, a nonlinear mathematical model has been proposed to study the dynamics of rabies infection among dogs and in a human population that is exposed to dog bites [C.S.Bornaa et al., 2020]. The model is shown to have a special, globally asymptotically stable disease-free equilibrium, and is also globally asymptotically stable whenever $R_0 \leq 1$

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Methodology

The following aspects were covered in developing the models presented in this study:

- Development of an seven-compartment fractional order model for the co-infection of bovine tuberculosis and rabies;
- Derivation of the Basic Reproduction Number for the three models;
- Calculation of the Equilibrium Points;
- Local and Global Stability Analyses of Equilibrium Points;
- Sensitivity Analysis using the Basic Reproduction Number;
- Optimal Model Formulation .

Concept and motivation of fractional order

We are familiar with $\dfrac{d^n y}{dx^n}, n \in \mathbb{N}$.
Let $y = x^n$, we know that
$y' = nx^{n-1}$

$y'' = n(n-1)x^{n-2} = \dfrac{n(n-1)(n-2)!}{(n-2)!}x^{n-2} = \dfrac{n!}{(n-2)!}x^{n-2}$

$y^{(3)} = \dfrac{n!}{(n-3)!}x^{n-3}$

For $k < n$, $\dfrac{d^k y}{dx^k} = \dfrac{n!}{(n-k)!}x^{n-k}$

Say, we want to calculate $\frac{1}{2}$ order, π^{th} order, etc.
$k \in \mathbb{N} \to \alpha \in \mathbb{R}$
$\dfrac{d^\alpha y}{dx^\alpha} = \dfrac{n!}{(n-\alpha)!}x^{n-\alpha}$ In 1695 hospital and leibniz.
In 1729 Euler developed Gamma-Euler function. $\Gamma(n) = \int_0^\infty e^{-t}t^{n-1}dt$
$\Gamma(n+1) = n!, n \in \mathbb{N}$ and $\Gamma(x+1) = x\Gamma(x), x \in \mathbb{R}$

$$\frac{d^\alpha y}{dx^\alpha} = \frac{n!}{\Gamma(n-\alpha+1)} x^{n-\alpha}$$

$$D^\alpha(y) = \frac{n!}{\Gamma(n-\alpha+1)} x^{n-\alpha}$$

Now we can calculate the half order derivative of $y = x$ with $n = 1, \alpha = \frac{1}{2}$

$$D^{\frac{1}{2}}(y) = \frac{1!}{\Gamma(1-\frac{1}{2}+1)} x^{1-\frac{1}{2}}$$

$$D^{\frac{1}{2}}(y) = \frac{2\sqrt{x}}{\sqrt{\pi}}$$

What about the constant? $y = cx^0, n = 0, \alpha = \frac{1}{2}$

$$D^{\frac{1}{2}}(c) = \frac{0!}{\Gamma(0-\frac{1}{2}+1)} x^{0-\frac{1}{2}}$$

$D^{\frac{1}{2}}(c) = \dfrac{1}{\sqrt{\pi x}}$ not cont at $x = 0$

Model Formulation on the transmission dynamics of co-infection of bovine TB and rabies

co-infection model assumptions

1. It is assumed that recruitment into the susceptible animal population is by birth and immigration at a constant rate.
2. Direct transmission among cattle follows standard incidence.
3. It is assumed that once cattle acquires bTB, they take some time before eventually showing clinical signs and symptoms.
4. The population in all compartments is positive.
5. Animals become infected by sucking the dairy product and the meat of infected cattle.

Model flow diagram

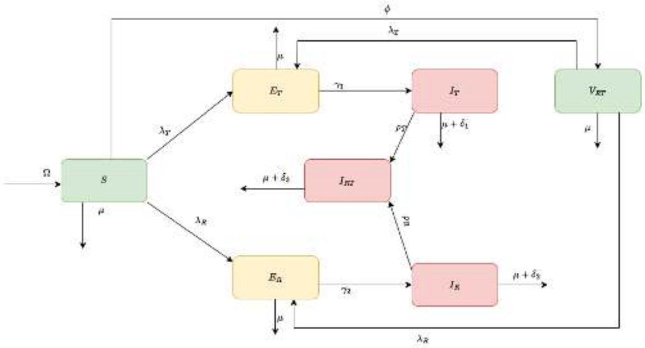

Figure 3: Schematic diagram for bovine TB and rabies transmission

Description of bovine TB Model Variables and Parameters

Table 1: Descriptions of co-infection Model Variables and Parameters

Symbols	Descriptions
S	Susceptible to both rabies and bovine TB
V	Vaccinated population from both diseases.
E_T	bovine TB exposed population
I_T	bovine TB infected population
E_R	Rabies exposed population
I_R	Rabies infected population
I_{RT}	Infected population to Rabies and bovine TB
Parameters	**Description**
Ω	Recruitment rate into the susceptible population
μ	Natural mortality rate at each stage
β_1	bovine TB transmission rate
β_2	Rabies transmission rate
γ_1	The proportion of bovine TB-exposed people who become infected.
γ_2	The proportion of Rabies-exposed people who become infected.
ϕ	Vaccination rate against bovine TB-Rabies infection
ε	Vaccine efficacy rate against bovine TB-Rabies infection
δ_1, δ_2, δ_3	Disease-induced death rate for animals in compartments I_T, I_R, and I_{RT}, respectively
ρ_T	The co-infection rate of rabies from bovine tuberculosis diseases.
ρ_R	The co-infection rate of bovine tuberculosis from rabies diseases.

Model formulation

From Figure 3, we have the following system of fractional differential equations:

$$
\begin{aligned}
{}_0^C D_t^{\alpha} S_H(t) &= (1-\kappa)\Lambda_H + \phi V_H - \lambda_H S_H - \mu_H S_H, \\
{}_0^C D_t^{\alpha} E_H(t) &= \lambda_H S_H + d\lambda_H V_H - (\mu_H + \sigma_H) E_H - \kappa E_H, \\
{}_0^C D_t^{\alpha} V_H(t) &= \kappa\Lambda_H + \kappa E_H - (\mu_H + \phi) V_H - d\lambda_H V_H, \\
{}_0^C D_t^{\alpha} I_H(t) &= \sigma_H E_H - (\mu_H + \gamma_H) I_H, \\
{}_0^C D_t^{\alpha} S_A(t) &= \Lambda_A - \lambda_A S_A - \mu_A S_A, \\
{}_0^C D_t^{\alpha} E_A(t) &= \lambda_A S_A - (\mu_A + \sigma_A) E_A, \\
{}_0^C D_t^{\alpha} I_A(t) &= \sigma_A E_A - (\mu_A + \gamma_A) I_A, \\
{}_0^C D_t^{\alpha} C_e(t) &= \rho I_A - \omega C_e
\end{aligned}
\tag{1}
$$

where ${}_0^C D^{\alpha}$ is the Caputo fractional derivative, $\lambda_H = \dfrac{\beta_1 I_H + \beta_2 I_A + \beta_3 C_e}{N_H}$, and $\lambda_A = \dfrac{\beta_4 I_H + \beta_5 I_A + \beta_6 C_e}{N_A}$

Model formlation

Numerous studies have taken into account the kind of mathematical formulation 1 [see, for example, [Ahmed et al., 2007, Area et al., 2015]. The fractional order model 1 clearly exhibits a mismatch in model dimensions, however Diethelm [Diethelm, 2013] has addressed this problem via fractionalization. He developed a fractional order compartmental model for dengue fever infection in his research article. Using statistics from the Cape Verde Islands' dengue disease epidemic, he showed how well the strategy worked. Using the approach of [Diethelm, 2013], [Sardar et al., 2015] proposed and examined a comparable model. They discovered that simulation results with data provide superior representation for epidemic models with memory in both vector and host populations than integer-order models.

Co-infection Fractional Differential Equations Model

$$
\begin{cases}
{}_0^C D_t^\alpha S(t) & = \Omega^\alpha - \left(\mu^\alpha + \phi^\alpha + \lambda_T^\alpha + \lambda_R^\alpha\right) S, \\
{}_0^C D_t^\alpha V(t) & = \phi^\alpha S - \left(\mu^\alpha + \varepsilon_1^\alpha \lambda_T^\alpha + \varepsilon_2^\alpha \lambda_R^\alpha\right) V \\
{}_0^C D_t^\alpha E_T(t) & = \lambda_T^\alpha S + \varepsilon_1^\alpha \lambda_T^\alpha V - \left(\mu^\alpha + \gamma_1^\alpha\right) E_T, \\
{}_0^C D_t^\alpha I_T(t) & = \gamma_1^\alpha E_T - \left(\mu^\alpha + \delta_1^\alpha + \rho_T^\alpha \lambda_R^\alpha\right) I_T, \\
{}_0^C D_t^\alpha E_R(t) & = \lambda_R^\alpha S + \varepsilon_2^\alpha \lambda_R^\alpha V - \left(\mu^\alpha + \gamma_2^\alpha\right) E_R, \\
{}_0^C D_t^\alpha I_R(t) & = \gamma_2^\alpha E_R - \left(\mu^\alpha + \delta_2^\alpha + \rho_R^\alpha \lambda_T^\alpha\right) I_R, \\
{}_0^C D_t^\alpha I_{RT}(t) & = \rho_T^\alpha \lambda_R^\alpha I_T + \rho_R^\alpha \lambda_T^\alpha I_R - (\mu^\alpha + \delta_3^\alpha) I_{RT}
\end{cases}
\tag{2}
$$

with $\quad S(t) > 0, V(t) > 0, E_T(t) > 0, I_T(t) > 0, E_R(t) > 0, I_R(t) > 0, I_{RT}(t) > 0 \quad \text{at} \quad t = 0$
where ${}_0^C D^\alpha$ is the Caputo fractional derivative.

Disease-Free Equilibrium (DFE) for co-infection model

The state in which the population is free from any diseases is referred to as the disease-free equilibrium. If both diseases are eliminated from the whole animal population, (Γ_1) indicates the disease-free equilibrium.

$$\Gamma_1 = \left(\frac{\Omega^\alpha}{\mu^\alpha + \phi^\alpha}, \frac{\phi^\alpha}{\mu^\alpha} \frac{\Omega^\alpha}{\mu^\alpha + \phi^\alpha}, 0, 0, 0, 0, 0 \right) \tag{3}$$

The Basic Reproduction Number of the model

The basic reproduction number (R_0) estimates the average number of new infections caused by an infected individual in a completely vulnerable community (see [Asamoah et al., 2017] and [Diekmann et al., 1990]). Using E_T, I_T, E_R, I_R and I_{RT} as infected compartments yields the following.

$$\mathcal{R}_0^T = \frac{\beta_1\Omega\,(\mu+(1-\varepsilon)\phi)\,(\mu+\gamma_1+\delta_1)}{\mu\,(\mu+\phi)\,(\mu+\gamma_1)\,(\mu+\delta_1)}, \tag{4}$$

which is the reproduction number only for bovine TB.

$$\mathcal{R}_0^R = \frac{\beta_2\Omega\,(\mu+(1-\varepsilon)\phi)\,(\mu+\gamma_2+\delta_2)}{\mu\,(\mu+\phi)\,(\mu+\gamma_2)\,(\mu+\delta_2)}, \tag{5}$$

which is the reproduction number only for rabies.

Mathematical analysis of the bovine Tuberculosis model

We look for invariant regions and evaluate the positivity of solutions to see if the model makes sense mathematically and epidemiologically. When the model's solutions are both positive and bounded, it becomes mathematically and biologically significant.

The model solutions' viability is demonstrated by the invariant area. We use the initials N_H and N_A to represent the human and animal groups of individuals, respectively, to examine the viability of the model solutions.

Mathematical analysis

Theorem 1:

Let $\Psi = \{(S_H(t), E_H(t), V_H(t), I_H(t), S_A(t), E_A(t), I_A(t), C_e(t)) \in \mathbb{R}_+^8 :$
$0 \leq N_H \leq \frac{\Lambda_H}{\mu_H} \cup 0 \leq N_A \leq \frac{\Lambda_A}{\mu_A} \cup 0 \leq C_e \leq \frac{\Lambda_A}{\mu_A}\frac{\rho}{\omega}\}$
The feasible solution set
$\{(S_H(t), E_H(t), V_H(t), I_H(t), S_A(t), E_A(t), I_A(t), C_e(t))\}$ of the system equation of the model enter and bounded in the region Ψ

Proof

To prove this, let us consider the human population, animal population, and contaminatted environment separately.

• The fractional derivative of the total human population, obtained by adding all the human equations of the model **??**, is given by

Proof

$$
\begin{aligned}
N_H(t) &= S_H(t) + E_H(t) + V_H(t) + I_H(t)\\
D^\alpha N_H &= D^\alpha S_H + D^\alpha E_H + D^\alpha V_H + D^\alpha I_H\\
D^\alpha N_H &= \Lambda_H^\alpha + \phi^\alpha V_H - \lambda_H^\alpha S_H - \kappa^\alpha S_H - \mu_H^\alpha S_H + \lambda_H^\alpha S_H - (\mu_H^\alpha + \sigma_H^\alpha)E_H\\
&\quad + \kappa^\alpha(S_H + E_H) - \mu_H^\alpha V_H + \sigma_H^\alpha E_H - (\mu_H^\alpha + \gamma_H^\alpha)I_H\\
D^\alpha N_H &= \Lambda_H^\alpha - \mu_H^\alpha S_H - \mu_H^\alpha E_H - \mu_H^\alpha V_H - \mu_H^\alpha I_H - \gamma_H^\alpha V_H - \gamma_H^\alpha I_H
\end{aligned}
$$

$$
D^\alpha N_H = \Lambda_H^\alpha - \mu_H^\alpha N_H - \gamma_H^\alpha I_H \tag{6}
$$

$$
D^\alpha N_H \le \Lambda_H^\alpha - \mu_H^\alpha N_H \tag{7}
$$

Proof

Let us take the Laplace transform[Apelblat, 2020] of equation 7 on both sides :

$$\mathcal{L}\left\{D_t^{\alpha} N_H(t)\right\}(s) + \mathcal{L}\left\{\mu_H N_H(t)\right\}(s) \geq \mathcal{L}\left\{\Lambda_H\right\}(s) \tag{8}$$

On the LHS:
$\mathcal{L}\left\{{}_aD_t^{\alpha} N_H(t)\right\}(s) = s^{\alpha}\mathcal{N}_H(t) - \sum_{k=0}^{n-1} s^{\alpha-k-1} N_H^{(k)}(0), \quad n-1 < \alpha \leq n$
Here $0 < \alpha < 1, \quad$ so $\quad n = 1,$
then, $\quad \mathcal{L}\left\{D_t^{\alpha} N_H(t)\right\}(s) = s^{\alpha}\mathcal{N}_H(s) - s^{\alpha-1} N_H(0)$, and

$$\mathcal{L}\left\{\mu_H N_H(t)\right\}(s) = \mu_H \mathcal{N}_H(s)$$

Proof

On the RHS

$$\mathcal{L}\left\{\Lambda_H\right\}(s) = \Lambda_H \mathcal{L}\{1\}$$
$$= \frac{\Lambda_H}{S}$$

Now the equation 8 becomes:

$$\mathcal{L}\left\{D_t^{\alpha} N_H(t)\right\}(s) + \mathcal{L}\left\{\mu_H N_H(t)\right\}(s) \geq \mathcal{L}\left\{\Lambda_H\right\}(s) \tag{9}$$
$$s^{\alpha}\mathcal{N}_H(s) - s^{\alpha-1}N_H(0) + \mu_H \mathcal{N}_H(s) \geq \frac{\Lambda_H}{S} \tag{10}$$
$$\mathcal{N}_H(s)(s^{\alpha} + \mu_H) \geq \frac{\Lambda_H}{S} + s^{\alpha-1}N_H(0) \tag{11}$$
$$\tag{12}$$

Hence, take $s^{\alpha-1}N_H(0) = 0$ at $t = 0$, [Tilahun et al., 2021] then

$$\mathcal{N}_H(s) \geq \Lambda_H \frac{s^{-1}}{s^{\alpha} + \mu_H} \tag{13}$$

Proof

Taking the inverse Laplace transform of $\mathcal{N}_H(s)$, and by using the Mittag-Leffler function, we have:

$$\begin{aligned} N_H(t) &\leq \Lambda_H \mathcal{L}^{-1}\left\{\frac{s^{-1}}{s^{\alpha}+\mu_H}\right\} \\ &\leq \Lambda_H t^{\alpha} E_{\alpha,\alpha+1}(-\mu_H t^{\alpha}) \\ &\leq \frac{\Lambda_H}{\mu_H}\left[1-E_{\alpha}(-\mu_H t^{\alpha})\right] \\ N_H(t) &\leq \frac{\Lambda_H}{\mu_H}\left[1-E_{\alpha}(-\mu_H t^{\alpha})\right] \end{aligned} \tag{14}$$

We have $\mu_H > 0$ and then, as $t \longrightarrow 0$, thus $N_H(t) \longrightarrow \frac{\Lambda_H}{\mu_H} \geq 0$. Therefore

$$0 \leq N_H(t) \leq \frac{\Lambda_H}{\mu_H} \tag{15}$$

$$\Psi_H = \left\{(S_H, E_H, V_H, I_H) \in \mathbb{R}_+^4 : S_H + E_H + V_H + I_H \leq \frac{\Lambda_H^{\alpha}}{\mu_H^{\alpha}}\right\}. \tag{16}$$

Proof

By the same approach, we'll get:

$$\Psi_A = \left\{ (S_A, E_A, I_A) \in \mathbb{R}_+^3 : S_H + E_H + I_H \leq \frac{\Lambda_A^\alpha}{\mu_A^\alpha} \right\} \tag{17}$$

for animal population, and

$$\Psi_{C_e} = \left\{ C_e \in \mathbb{R}_+ : C_e \leq \frac{\Lambda_A^\alpha}{\mu_A^\alpha} \frac{\rho^\alpha}{\omega^\alpha} \right\} \tag{18}$$

For contaminated environment

feasible region

The feasible region for the system of fractional differential equations in **??** is given by:

$$\Psi = \Psi_H \times \Psi_A \times \Psi_{C_e} \subset \mathbb{R}_+^4 \times \mathbb{R}_+^3 \times \mathbb{R}_+, \tag{19}$$

which is a positive invariant set.
This shows the boundedness of the solution of the model. □

Positivity of the solution

In this section, we showed all the solution of the models Equation **??** remains positive for future time if their respective initial values are positive. To establish this second result, we introduce the following lemma.

Lemma

(Generalized Mean Value Theorem) [Kheiri and Jafari, 2018]
Suppose that $z(t) \in C[a, b]$ and ${}_0^C D_t^\alpha z(t) \in C[a, b]$ for $0 < \alpha \leq 1$, then

$$z(t) = z(a) + \frac{1}{\Gamma(\alpha)} {}_0^C D_t^\alpha z(\eta).(t-a)^\alpha, \tag{20}$$

where $a \leq \eta \leq t, \forall t \in (a, b]$.

Remark

Assume that $z(t) \in C[a,b]$ and ${}_0^C D_t^\alpha z(t) \in C[a,b]$ for $0 < \alpha \leq 1$. It follows from Lemma 5.1 that if ${}_0^C D_t^\alpha z(t) \geq 0, \forall t \in (a,b)$, then $z(t)$ is increasing for $\forall t \in [a,b]$, and if ${}_0^C D_t^\alpha z(t) \leq 0, \forall t \in (a,b)$ then $z(t)$ is decreasing for $\forall t \in [a,b]$

Theorem

If $S_H(0), E_H(0), V_H(0), I_H(0), S_A(0), E_A(0), I_A(0), C_e(0)$ are positives, then $S_H(t), E_H(t), V_H(t), I_H(t), S_A(t), E_A(t), I_A(t), C_e(t)$ are also positives for all time $t > 0$;

Proof

Let us take all the equations of the model in Equation **??** at each variable equal to zero, we have:

$$ {}_0^C D_t^\alpha S_H|_{S_H=0} = (1-\kappa^\alpha)\Lambda_H^\alpha + \phi^\alpha V_H \geq 0 \quad (21)$$
$$ {}_0^C D_t^\alpha E_H|_{E_H=0} = \lambda_H S_H + d\lambda_H V_H \geq 0 \quad (22)$$
$$ {}_0^C D_t^\alpha V_H|_{V_H=0} = \kappa^\alpha \Lambda_H^\alpha + \kappa^\alpha E_H \geq 0 \quad (23)$$
$$ {}_0^C D_t^\alpha I_H|_{I_H=0} = \sigma_H^\alpha E_H \geq 0 \quad (24)$$
$$ {}_0^C D_t^\alpha S_A|_{S_A=0} = \Lambda_A^\alpha > 0 \quad (25)$$
$$ {}_0^C D_t^\alpha E_A|_{E_A=0} = \lambda_A S_A \geq 0 \quad (26)$$
$$ {}_0^C D_t^\alpha I_A|_{I_A=0} = \sigma_A^\alpha E_A \geq 0 \quad (27)$$
$$ {}_0^C D_t^\alpha C_e|_{C_e=0} = \rho^\alpha I_A \geq 0 \quad (28)$$

Proof

Since $S_H(0), E_H(0), V_H(0), I_H(0), S_A(0), E_A(0), I_A(0), C_e(0)$ are positives, according to equations 21 - 28 and the remark 5.1, the solution $(S_H(t), E_H(t), V_H(t), I_H(t), S_A(t), E_A(t), I_A(t), C_e(t))$ can't scape from the hyperplanes of $S_H = 0, E_H = 0, V_H = 0, V_H = 0, I_H = 0, S_A = 0$ $E_A = 0, I_A = 0$, and $C_e = 0$ Therefore, all the solutions of the model with initial conditions in Ψ remain in Ψ for all $t > 0$. Thus, this region is a positive invariant set.
The model **??** is mathematically and epidemiologically meaningful; therefore, we can consider the flow generated by the model for analysis. $\square$

Disease-Free Equilibrium (DFE)

The condition when there are no diseases in the population is known as the disease-free equilibrium point. When bTB is absent from both human and animal populations, Φ_0 provides the disease-free equilibrium.

$$\begin{cases} {}_0^C D_t^\alpha S_H(t) &= 0, \\ {}_0^C D_t^\alpha E_H(t) &= 0, \\ {}_0^C D_t^\alpha V_H(t) &= 0, \\ {}_0^C D_t^\alpha I_H(t) &= 0, \\ {}_0^C D_t^\alpha S_A(t) &= 0, \\ {}_0^C D_t^\alpha E_A(t) &= 0, \\ {}_0^C D_t^\alpha I_A(t) &= 0, \\ {}_0^C D_t^\alpha C_e(t) &= 0 \end{cases} \tag{29}$$

Disease-Free Equilibrium (DFE)

After some calculus we get the desease free equilibruim:

$$\Phi_0 = \left(\frac{\Lambda_H^\alpha \left(\phi^\alpha + (1-\kappa^\alpha)\mu_H^\alpha\right)}{\mu_H(\mu_H^\alpha + \phi^\alpha)}, 0, \frac{\kappa^\alpha \Lambda_H^\alpha}{\mu_H^\alpha + \phi^\alpha}, 0, \frac{\Lambda_A^\alpha}{\mu_A^\alpha}, 0, 0, 0 \right) \tag{30}$$

The Basic Reproduction Number

The basic reproduction number R_0 describes the typical number of new cases that a single infectious person creates when they are introduced into a community that is completely susceptible [Diekmann et al., 1990].

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The Basic Reproduction Number

The basic reproduction number R_0 describes the typical number of new cases that a single infectious person creates when they are introduced into a community that is completely susceptible [Diekmann et al., 1990].

The Basic Reproduction Number cont'

Next-generation matrix

We employ the next-generation matrix technique, taking new infections and transfer terms into account, to get the basic number for reproduction R_0 [Diekmann et al., 1990, Van den Driessche and Watmough, 2002].
The basic reproduction number R_0 is expressed as the greatest eigenvalue if the new infectious and transfer terms for bTB are indicated by F_i and V_i, respectively.
We have,

$$R_0 = \rho(FV^{-1})$$

where

$$F = \left| \frac{\partial F_i x(0)}{\partial x_j} \right|, \quad V = \left| \frac{\partial V_i x(0)}{\partial x_j} \right|,$$

ρ denotes here the spectral radius of a matrix which is the greatest eigenvalue of a given matrix.

The Basic Reproduction Number cont'd

$$R_1 = \frac{\Lambda_H \beta_1 \sigma_H \left(d\kappa\,\mu_H - \kappa\,\mu_H + \phi + \mu_H\right)}{\mu_H\left(\mu_H + \phi\right)\left(\mu_H + \sigma_H + \kappa\right)\left(\mu_H + \gamma_H\right)} + \frac{\Lambda_A \sigma_A (\beta_5 \omega + \beta_6 \rho)}{\mu_A\left(\mu_A + \sigma_A\right)\left(\mu_A + \gamma_A\right)\omega} \tag{31}$$

$$R_2 = \frac{\Lambda_H \beta_1 \sigma_H \left(d\kappa\,\mu_H - \kappa\,\mu_H + \phi + \mu_H\right)}{\mu_H\left(\mu_H + \phi\right)\left(\mu_H + \sigma_H + \kappa\right)\left(\mu_H + \gamma_H\right)} - \frac{\Lambda_A \sigma_A (\beta_5 \omega + \beta_6 \rho)}{\mu_A\left(\mu_A + \sigma_A\right)\left(\mu_A + \gamma_A\right)\omega} \tag{32}$$

$$R_3 = \frac{\beta_4 \Lambda_A \sigma_H \Lambda_H \sigma_A \left(d\kappa\,\mu_H - \kappa\,\mu_H + \phi + \mu_H\right)\left(\omega\,\beta_2 + \rho\,\beta_3\right)}{\omega \mu_A \mu_H \left(\mu_H + \phi\right)\left(\mu_A + \sigma_A\right)\left(\mu_A + \gamma_A\right)\left(\mu_H + \sigma_H + \kappa\right)\left(\mu_H + \gamma_H\right)} \tag{33}$$

$$R_0 = \frac{R_1}{2} + \frac{\sqrt{R_2^2 + 4R_3}}{2} \tag{34}$$

The Basic Reproduction Number cont'd

Therefore, Equation (35) provides the fundamental reproduction number (R_0) of the entire model (2).

$$\mathcal{R}_0 = max(\mathcal{R}_0^T, \mathcal{R}_0^R) \tag{35}$$

Remark

$\mathcal{R}_0$ denotes the secondary infection resulting from the infectious compartment of animals following the vaccination, treatment or quarantine, and the culling of exposed animals. If $\mathcal{R}_0 < 1$, the infection will eventually be eradicated from animals. In contrast, if $\mathcal{R}_0 > 1$, the illness persists in animals for a prolonged period, increasing the risk of infection in susceptible animals.

Variables

The list of variables and parameters used are as below

Symbol	Definition
S_H	Susceptible human population
E_H	Exposed human population
I_H	Infected human population
V_H	Vaccinated human population
S_C	Susceptible canine population
E_C	Exposed canine population
I_C	Infected canine population
V_C	Vaccinated canine population

Table 2: Rabies model Variables and their defintions

Impact of bovine Tuberculosis on Rabies

Bovine tuberculosis can manifest as subacute or chronic, with varying rates of progression. Some animals may develop serious illness within months after infection, while others may take several years to show clinical symptoms. The bacteria can also remain dormant in the host, so no disease can be caused for an extended period of time. Although most pets show rabies symptoms within three weeks to three months after virus exposure, symptoms can sometimes take months, or even more than a year, to appear. Therefore, this part investigates only how bovine tuberculosis affects rabies, and not the reverse [WHO, 2016]. This is done by expressing the reproduction number of bovine TB in terms of rabies.

Impact of bovine Tuberculosis on Rabies

Let us begin by expressing (Ω) (which is typical for both equilibrium points) in equation (4)in terms of $\mathcal{R}_0^T$.

$$\mathcal{R}_0^T = \frac{\beta_1 \Omega \left(\mu + (1-\varepsilon)\phi\right)\left(\mu + \gamma_1 + \delta_1\right)}{\mu\left(\mu+\phi\right)\left(\mu+\gamma_1\right)\left(\mu+\delta_1\right)},$$

solving for (Ω) gives

$$(\Omega) = \frac{\mathcal{R}_0^T \mu\left(\mu+\phi\right)\left(\mu+\gamma_1\right)\left(\mu+\delta_1\right)}{\beta_1 \left(\mu + (1-\varepsilon)\phi\right)\left(\mu + \gamma_1 + \delta_1\right)}$$

By replacing (Ω) in $\mathcal{R}_0^R$ according to equation (5), we express $\mathcal{R}_0^R$ in terms of $\mathcal{R}_0^T$
as

$$\mathcal{R}_0^R = \frac{\beta_2 \left(\mu+\gamma_1\right)\left(\mu+\delta_1\right)}{\beta_1 \left(\mu+\delta_1+\gamma_1\right)} \frac{\left(\mu+\delta_2+\gamma_2\right)}{\left(\mu+\gamma_2\right)\left(\mu+\delta_2\right)} \mathcal{R}_0^T$$

Impact of bovine Tuberculosis on Rabies

The rate of change of $\mathcal{R}_0^R$ with respect to $\mathcal{R}_0^T$ is represented by the following partial derivative:

$$\frac{\partial \mathcal{R}_0^R}{\partial \mathcal{R}_0^T} = \frac{\beta_2\ (\mu+\gamma_1)(\mu+\delta_1)(\mu+\delta_2+\gamma_2)}{\beta_1\ (\mu+\gamma_2)(\mu+\delta_2)(\mu+\delta_1+\gamma_1)}$$

We have determined that the partial derivative of $\mathcal{R}_0^R$ with respect to $\mathcal{R}_0^T$ is positive. This finding suggests that an increase in bovine TB infections within the population will increase the transmission of rabies.

Sensitivity Analysis of Co-infection model

Table 3: Summary of the parameters and their corresponding values for the co-infection model.

Parameter	Value	Sources
Ω	50	[Shirima Sabini et al., 2020]
μ	0.1	[Asamoah et al., 2017]
β_1	0.56e-1	[Shirima Sabini et al., 2020]
β_2	0.01	[Asamoah et al., 2017]
γ_1	0.350	Given
γ_2	0.58	Given
δ_1	$2.37/6$	[Asamoah et al., 2017]
δ_2	0.85	[Asamoah et al., 2017]
δ_3	0.1	Given
ϕ	0.3	Given
ε	0.66	[Asamoah et al., 2017]

Examination of the Rabies Sub-Model Sensitivity

Table 4: The sensitivity indices for the exclusive Rabies sub-model are derived by substituting the parameter values from Table 3.

Parameter	Description	Sensitivity indices
Ω	Recruitment rate into the susceptible population	$+1$
μ	Natural mortality rate at each stage	-0.9419
β_2	Rabies transmission rate	$+1$
γ_2	The proportion of Rabies-exposed individuals who become infected.	-0.4274
δ_2	Disease-induced death rate for animals in compartment I_R	-0.3483
ϕ	Vaccination rate against bovine TB-Rabies infection	-0.2450
ε	Vaccine efficacy rate against bovine TB-Rabies infection	-0.9802

The analysis of sensitivity for the basic reproduction number related to the Rabies submodel parameters, as provided from the rabies sub-model, employing the normalized forward sensitivity index of the basic reproduction number, is presented as follows: $\frac{\partial \mathcal{R}_0^R}{\partial \epsilon} \frac{\epsilon}{\mathcal{R}_0^T}$, ϵ represents any parameter in $\mathcal{R}_0^R$.

Examination of the bovine TB Sub-Model Sensitivity

Table 5: The sensitivity indices for the exclusive Rabies sub-model are derived by substituting the parameter values from Table 3.

Parameter	Description	Sensitivity indices
Ω	Recruitment rate into the susceptible population	$+1$
μ	Natural mortality rate at each stage	-0.9419
β_1	Bovine TB transmission rate	$+1$
γ_1	The proportion of bovine TB-exposed individuals who become infected.	-0.4274
δ_1	Disease-induced death rate for animals in compartment I_T	-0.3483
ϕ	Vaccination rate against bovine TB-Rabies infection	-0.2450
ε	Vaccine efficacy rate against bovine TB-Rabies infection	-0.9802

Referring to Table of sensitivity indices 5, it is evident that the contact rate of bovine tuberculosis β_1 along with the recruitment rate Ω significantly promotes the spread of the disease. In contrast, parameters such as μ, ε, δ_1, γ_1, and ϕ exhibit a negative influence, indicating that their increase would lead to a reduction in the number of people infected with bovine tuberculosis.

Fractional Optimal Control of the Co-infection Model

To assess the effects of intervention strategies, the model incorporates time-varying controls $u_1(t)$, $u_2(t)$, and $u_3(t)$ as outlined in (2).
Below are the defined controls:

- i. $u_1(t)$: Control aimed at preventing bovine tuberculosis and rabies by implementing a public health education campaign and concentrating efforts on vaccinating cattle, with the goal of reducing infections from contaminated environments and inter-cattle transmissions to eventually eliminate the disease within a localized region,
- ii. $u_2(t)$: involves the slaughter of cattle infected with bovine tuberculosis to minimize the risk of spreading the disease from these infected animals,
- iii. $u_3(t)$: involves the slaughter of cattle infected with rabies to minimize the risk of spreading the disease from these infected animals.

The system of differential equations resulting from the incorporation of these controls into the model is as follows:

$$
\begin{cases}
{}_0^C D_t^\alpha S(t) & = \Omega^\alpha - (\mu^\alpha + u_1\phi^\alpha + \lambda_T^\alpha + \lambda_R^\alpha)\, S, \\
{}_0^C D_t^\alpha V(t) & = u_1\phi^\alpha S - (\mu^\alpha + (1-u_1)\varepsilon_1^\alpha\lambda_T^\alpha + (1-u_1)\varepsilon_2^\alpha\lambda_R^\alpha)\, V, \\
{}_0^C D_t^\alpha E_T(t) & = \lambda_T^\alpha S + (1-u_1)\varepsilon_1^\alpha\lambda_T^\alpha V - (\mu^\alpha + \gamma_1^\alpha)\, E_T, \\
{}_0^C D_t^\alpha I_T(t) & = \gamma_1^\alpha E_T - (\mu^\alpha + \delta_1^\alpha + u_2 + \rho_T^\alpha\lambda_R^\alpha)\, I_T, \\
{}_0^C D_t^\alpha E_R(t) & = \lambda_R^\alpha S + (1-u_1)\varepsilon_2^\alpha\lambda_R^\alpha V - (\mu^\alpha + \gamma_2^\alpha)\, E_R, \\
{}_0^C D_t^\alpha I_R(t) & = \gamma_2^\alpha E_R - (\mu^\alpha + \delta_2^\alpha + u_3 + \rho_R^\alpha\lambda_T^\alpha)\, I_R, \\
{}_0^C D_t^\alpha I_{RT}(t) & = \rho_T^\alpha\lambda_R^\alpha I_T + \rho_R^\alpha\lambda_T^\alpha I_R - (\mu^\alpha + \delta_3^\alpha) I_{RT}
\end{cases}
\tag{36}
$$

with $\quad S(t) > 0, V(t) > 0, E_T(t) > 0, I_T(t) > 0, E_R(t) > 0, I_R(t) > 0, I_{RT}(t) > 0 \quad \text{at} \quad t = 0$

We investigate the optimal level of control efforts to minimize the number of bovine TB-infected animals in the asymptomatique stage (E_T), the number of rabies-infected animals in the asymptomatique stage (E_R), the number of bovine tuberculosis rabies co-infected animals I_{RT}, and also the cost of applying the controls. This is done by minimizing the following objective functional:

$$j(u) = \int_0^b \left[A1 I_T + A2 I_R + A3 I_{RT} + \frac{B1}{2} u_1^2(t) + \frac{B2}{2} u_2^2(t) + \frac{B3}{2} u_3^2(t) \right] dt, \tag{37}$$

Subject to (36).

A1, A2, A3, B1, B2, and B3 are positive weights, where A1, A2 and A3 are constant weights of animals infected to bovine TB, animals infected to rabies, and rabies and bovine TB co-infected animals, respectively. B1, B2, and B3 are weights that balance the cost factors associated with the control strategies.

Necessary Conditions for FOCP Optimality

The FOCP in question is defined as follows: Determine the optimal control $u(t)$ that reduces the performance index

$$j(u) = \int_0^b f(t, x, u)dt, \tag{38}$$

subject to the dynamic constraint

$$^{C}_{0}D^{\alpha}_{t}x(t) = g(t, x, u), \tag{39}$$

with initial condition

$$x(0) = x_0 \tag{40}$$

where $x(t)$ and $u(t)$ are the state and control variables, respectively,f and g are differentiable functions and $0 < \alpha \leq 1$.

Necessary Conditions for FOCP Optimality

Theorem

If (x, u) is a minimizer of (38) under dynamic constraints (39) and the boundary condition (40), then there exists a function $\lambda \in C^1[0, b]$ such that the triplet (x, u, λ) satisfies:

- The state and co-state systems;

$$ {}_0^C D_t^\alpha x(t) = \frac{\partial H}{\partial \lambda} = (t, x(t), u(t), \lambda(t)), \tag{41} $$

$$ {}_t^C D_b^\alpha \lambda(t) = \frac{\partial H}{\partial x} = (t, x(t), u(t), \lambda(t)), \tag{42} $$

- the stationary condition

$$ \frac{\partial H}{\partial u} = (t, x(t), u(t), \lambda(t)) = 0 \tag{43} $$

- and the transversality condition

$$ \lambda(b) = 0 \tag{44} $$

where H is the Hamiltonian, and it is defined by:

$$ H(t, x, u, \lambda) = f(t, x, u) + \lambda \cdot g(t, x, u). \tag{45} $$

Proof.

A proof of this theorem is given in [Kheiri and Jafari, 2018] □

Defining the Optimal Control

we establish the necessary conditions for an optimal control and develop an optimality system that describes the optimal control by employing the upper and lower-bound method. The optimal control triplet values can be calculated using the optimality system. To achieve this, we start by expressing the Hamiltonian like in [Agrawal and Baleanu, 2007] and [Kheiri and Jafari, 2018] of the problem (37) as follows:

$$
\begin{aligned}
H = & \left[A1 I_T + A2 I_R + A3 I_{RT} + \frac{B1}{2} u_1^2(t) + \frac{B2}{2} u_2^2(t) + \frac{B3}{2} u_3^2(t)\right] \\
& + \lambda_S \left[\Omega^\alpha - \left(\mu^\alpha + u_1 \phi^\alpha + \lambda_T^\alpha + \lambda_R^\alpha\right) S\right] \\
& + \lambda_V \left[u_1 \phi^\alpha S - \left(\mu^\alpha + (1-u_1)\varepsilon_1^\alpha \lambda_T^\alpha + (1-u_1)\varepsilon_2^\alpha \lambda_R^\alpha\right) V\right] \\
& + \lambda_{E_T} \left[\lambda_T^\alpha S + (1-u_1)\varepsilon_1^\alpha \lambda_T^\alpha V - \left(\mu^\alpha + \gamma_1^\alpha\right) E_T\right] \\
& + \lambda_{I_T} \left[\gamma_1^\alpha E_T - \left(\mu^\alpha + \delta_1^\alpha + u_2 + \rho_T^\alpha \lambda_R^\alpha\right) I_T\right] \\
& + \lambda_{E_R} \left[\lambda_R^\alpha S + (1-u_1)\varepsilon_2^\alpha \lambda_R^\alpha V - \left(\mu^\alpha + \gamma_2^\alpha\right) E_R\right] \\
& + \lambda_{I_R} \left[\gamma_2^\alpha E_R - \left(\mu^\alpha + \delta_2^\alpha + u_3 + \rho_R^\alpha \lambda_T^\alpha\right) I_R\right] \\
& + \lambda_{I_{RT}} \left[\rho_T^\alpha \lambda_R^\alpha I_T + \rho_R^\alpha \lambda_T^\alpha I_R - (\mu^\alpha + \delta_3^\alpha) I_{RT}\right]
\end{aligned}
\tag{46}
$$

Where λ_S, λ_V, λ_{E_T}, λ_{I_T}, λ_{E_R}, λ_{I_R} and $\lambda_{I_{RT}}$ are adjoint or co-state variables corresponding to the state variables S, V, E_T, I_T, E_R, I_R and I_{RT}, respectively.

Theorem

Assume $u^* = (u_1^*, u_2^*, u_3^*)$ is an optimal control triplet for (37) over U, with associated optimal states S^{u^*}, V^{u^*}, $E_T^{u^*}$, $I_T^{u^*}$, $E_R^{u^*}$, $I_R^{u^*}$, and $I_{RT}^{u^*}$ of the system (36) that minimizes (37) over U. Consequently, there exist adjoint functions λ_S, λ_V, λ_{E_T}, λ_{I_T}, λ_{E_R}, λ_{I_R}, and $\lambda_{I_{RT}}$ that satisfy the following.

Necessary conditions for an optimal control

$$
\begin{aligned}
{}_0^C D_t^\alpha \lambda_S(b-t) &= -\mu\lambda_S + u_1\phi\left[\lambda_V(b-t) - \lambda_S(b-t)\right] + \lambda_T\left[\lambda_{E_T}(b-t) - \lambda_S(b-t)\right] \\
&\quad + \lambda_R\left[\lambda_{E_R}(b-t) - \lambda_S(b-t)\right] \\
{}_0^C D_t^\alpha \lambda_V(b-t) &= -\mu\lambda_V + \varepsilon_1\lambda_T\left[\lambda_{E_T}(b-t) - \lambda_V(b-t)\right] + \varepsilon_2\lambda_R\left[\lambda_{E_R}(b-t) - \lambda_V(b-t)\right] \\
{}_0^C D_t^\alpha \lambda_{E_T}(b-t) &= -\mu\lambda_{E_T} + \beta_1 S\left[\lambda_{E_T}(b-t) - \lambda_S(b-t)\right] + \varepsilon_1\beta_1 V\left[\lambda_{E_T}(b-t) - \lambda_V(b-t)\right] \\
&\quad + \gamma_1\left[\lambda_{I_T}(b-t) - \lambda_{E_T}(b-t)\right] + \rho_R\beta_1 I_R\left[\lambda_{I_{RT}}(b-t) - \lambda_{I_R}(b-t)\right] \\
{}_0^C D_t^\alpha \lambda_{I_T}(b-t) &= A1 + \beta_1 S\left[\lambda_{E_T}(b-t) - \lambda_S(b-t)\right] + \varepsilon_1\beta_1 V\left[\lambda_{E_T}(b-t) - \lambda_V(b-t)\right] - (\mu+\delta_1+u_2) \\
&\quad \lambda_{I_T}(b-t) + \rho_T\lambda_R\left[\lambda_{I_{RT}}(b-t) - \lambda_{I_T}(b-t)\right] + \rho_R\beta_1 I_R\left[\lambda_{I_{RT}}(b-t) - \lambda_{I_R}(b-t)\right] \\
{}_0^C D_t^\alpha \lambda_{E_R}(b-t) &= -\mu\lambda_{E_R} + \beta_2 S\left[\lambda_{E_R}(b-t) - \lambda_S(b-t)\right] + \varepsilon_2\beta_2 V\left[\lambda_{E_R}(b-t) - \lambda_V(b-t)\right] \\
&\quad + \gamma_2\left[\lambda_{I_R}(b-t) - \lambda_{E_R}(b-t)\right] + \rho_T\beta_2 I_T\left[\lambda_{I_{RT}}(b-t) - \lambda_{I_T}(b-t)\right] \\
{}_0^C D_t^\alpha \lambda_{I_R}(b-t) &= A2 + \beta_2 S\left[\lambda_{E_R}(b-t) - \lambda_S(b-t)\right] + \varepsilon_2\beta_2 V\left[\lambda_{E_R}(b-t) - \lambda_V(b-t)\right] - (\mu+\delta_2+u_3) \\
&\quad \lambda_{I_R}(b-t) + \rho_R\lambda_T\left[\lambda_{I_{RT}}(b-t) - \lambda_{I_R}(b-t)\right] + \rho_T\beta_2 I_T\left[\lambda_{I_{RT}}(b-t) - \lambda_{I_T}(b-t)\right] \\
{}_0^C D_t^\alpha \lambda_{I_{RT}}(b-t) &= A3 + \beta_1 S\left[\lambda_{E_T}(b-t) - \lambda_S(b-t)\right] + \beta_2 S\left[\lambda_{E_R}(b-t) - \lambda_S(b-t)\right] \\
&\quad + \varepsilon_1\beta_1 V\left[\lambda_{E_T}(b-t) - \lambda_V(b-t)\right] + (1-u_1)\varepsilon_2\beta_2 V\left[\lambda_{E_R}(b-t) - \lambda_V(b-t)\right] \\
&\quad + \rho_T\beta_2 I_T\left[\lambda_{I_{RT}}(b-t) - \lambda_{I_T}(b-t)\right] + \rho_R\beta_1 I_R\left[\lambda_{I_{RT}}(b-t) - \lambda_{I_R}(b-t)\right] \\
&\quad - (\mu+\delta_3)\lambda_{I_{RT}}(b-t)
\end{aligned}
\tag{47}
$$

under the subsequent initial conditions:

$$S(0) = S_0, V(0) = V_0, E_T(0) = E_{T_0}, I_T(0) = I_{T_0}, E_R(0) = E_{R_0}, I_R(0) = I_{R_0}, I_{RT}(0) = I_{RT_0}, \tag{48}$$

with transversality conditions

$$\lambda_S(b) = \lambda_V(b) = \lambda_{E_T}(b) = \lambda_{I_T}(b) = \lambda_{E_R}(b) = \lambda_{I_R}(b) = \lambda_{I_{RT}}(b) = 0 \tag{49}$$

Furthermore, the optimal controls u_1^*, u_2^* and u_3^* are given by:

$$u_1^* = \min\left(1, \max\left(0, \frac{\lambda_T S\left(\lambda_{E_T} - \lambda_S\right)}{B1}\right)\right), \tag{50}$$

$$u_2^* = \min\left(1, \max\left(0, \frac{\lambda_R S\left(\lambda_{E_R} - \lambda_S\right)}{B2}\right)\right), \tag{51}$$

$$u_3^* = \min\left(1, \max\left(0, \frac{\phi S\left(\lambda_S - \lambda_V\right)}{B3}\right)\right), \tag{52}$$

Numerical Simulation and discussion

Numerical Simulation

The numerical simulation of the model was done using the Predict-Evaluate-Correct-Evaluate (PECE) method of Adam-Bashforth-Moulton in MATLAB.

Advantages and Applications

- **Increased Accuracy**: By using both predictor and corrector steps, the method reduces error.
- **Stability**: The implicit nature of the corrector step can improve stability, especially for stiff equations.
- **Efficiency**: Computationally efficient for solving large systems of ODEs.

Simulation Results for the Optimal Control Problem of Co-infection model

The numerical simulations are performed using Matlab software with parameters listed in Table 3. The weight coefficients are $A1 = 20$, $A2 = 2$, $A3 = 10$, $B1 = 200$, $B2 = 100$, $B3 = 50$. The initial conditions are $S(0) = 150$, $V(0) = 5$, $E_T(0) = 10$, $I_T(0) = 3$, $E_R(0) = 3$, $I_R(0) = 1$, $I_{RT}(0) = 1$, and various values for the derivative order α where $0.75 \leq \alpha \leq 1$.

Differents strategies

1. **Strategy A**: Implementing vaccination measures exclusively (i.e. $u_1 \neq 0$,$u_2 = 0$, and $u_3 = 0$).
2. **Strategy B**: Applying vaccination and removal control for animals affected by rabies (i.e. $u_1 \neq 0$, $u_2 = 0$, and $u_3 \neq 0$).
3. **Strategy C**: Implementing culling measures for animals that have contracted rabies and those infected with bovine TB (i.e. $u_1 = 0$, $u_2 \neq 0$, and $u_3 \neq 0$).
4. **Strategy D**: Implementing only removal control for animals affected by rabies (i.e. $u_1 = 0$, $u_2 = 0$ and $u_3 \neq 0$).
5. **Strategy E**: Implementing all controls (i.e. $u_1 \neq 0$, $u_2 \neq 0$ and $u_3 \neq 0$).

Strategy A: Implementing the vaccination measure exclusively

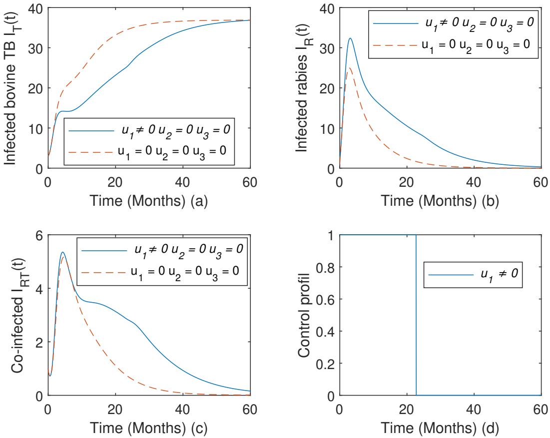

Figure 4: Effects of vaccination control on bovine TB, rabies and the co-infection transmission dynamics, with $\alpha = 0.995$

Strategy A: Implementing the vaccination measure exclusively Cont'd

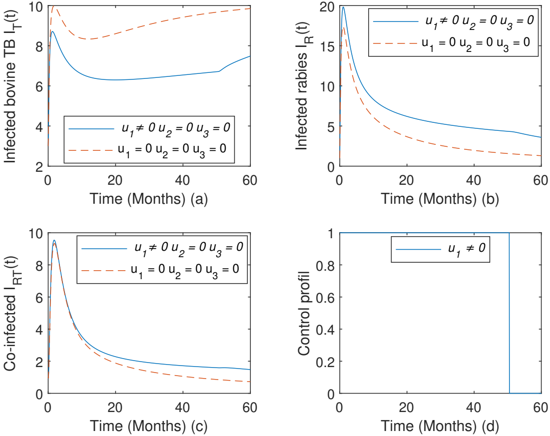

Figure 5: Effects of vaccination control on bovine TB, rabies and the co-infection transmission dynamics, with $\alpha = 0.75$

Strategy B: Applying vaccination and removal control for animals affected by rabies

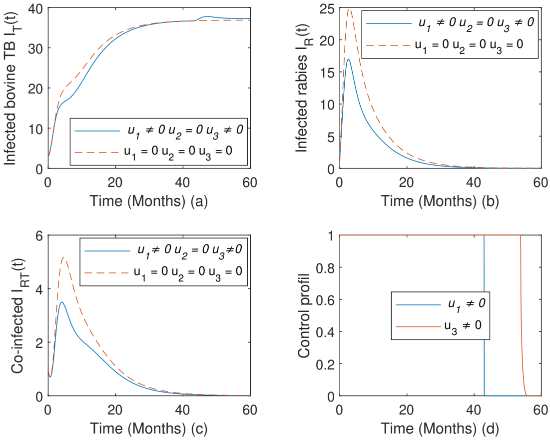

Figure 6: Impact of vaccination and infected rabies animal culling on the transmission dynamics of bovine tuberculosis, rabies, and co-infection, given

Strategy B: Applying vaccination and removal control for animals affected by rabies Cont'd

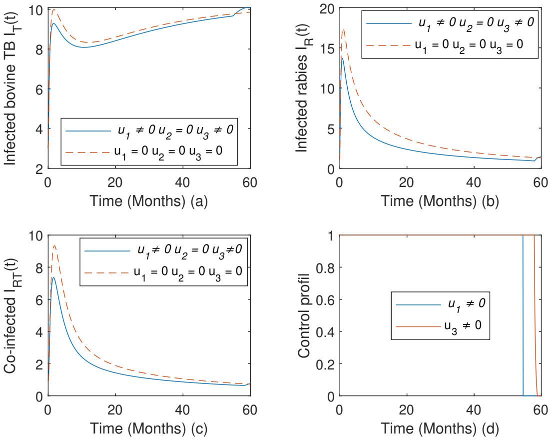

Figure 7: Impact of vaccination and infected rabies animal culling on the transmission dynamics of bovine tuberculosis, rabies, and co-infection, given

Strategy C: Implementing culling measures for animals that have contracted rabies and those infected with bovine TB

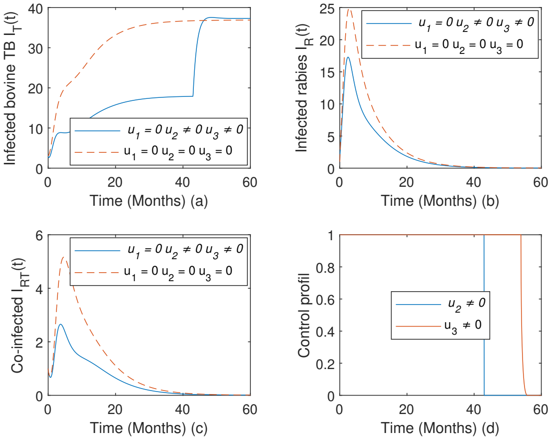

Figure 8: Effects culling controls for animals that have contracted rabies and those infected with bovine TB, with $\alpha = 0.995$

Strategy C: Implementing culling measures for animals that have contracted rabies and those infected with bovine TB Cont‘d

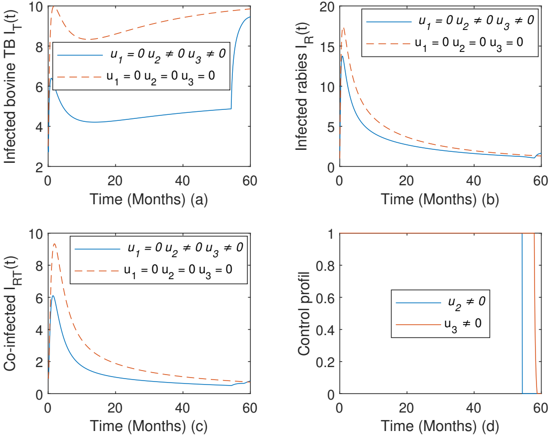

Figure 9: Effects culling controls for animals that have contracted rabies and

Strategy D: Implementing only removal control for animals affected by rabies

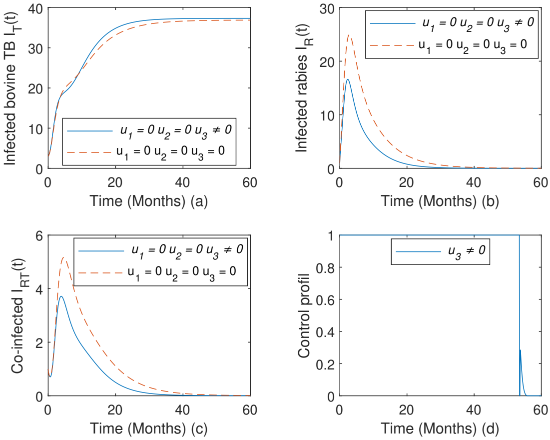

Figure 10: Impacts of implementing a single culling measure in rabies-infected animals, with $\alpha = 0.995$

Strategy D: Implementing only removal control for animals affected by rabies Cont'd

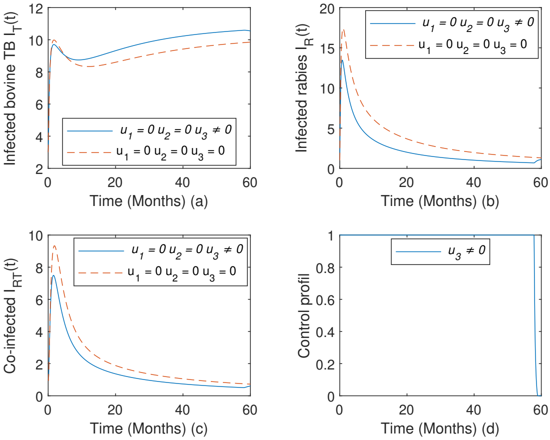

Figure 11: Impacts of implementing a single culling measure in rabies-infected animals, with $\alpha = 0.75$

Strategy E: Implementing of all controls

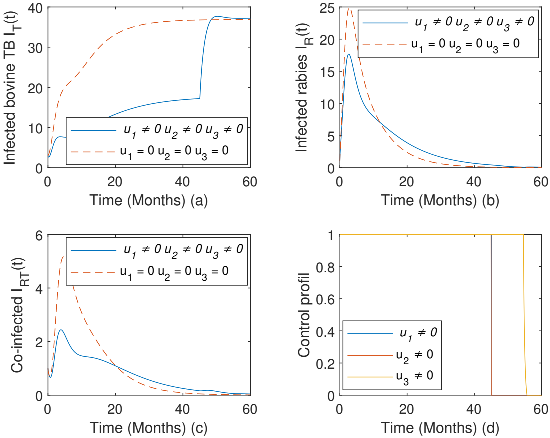

Figure 12: Effects of all controls on bovine TB, rabies and their co-infection transmission dynamics, with $\alpha = 0.995$

Strategy E: Implementing of all controls Cont'd

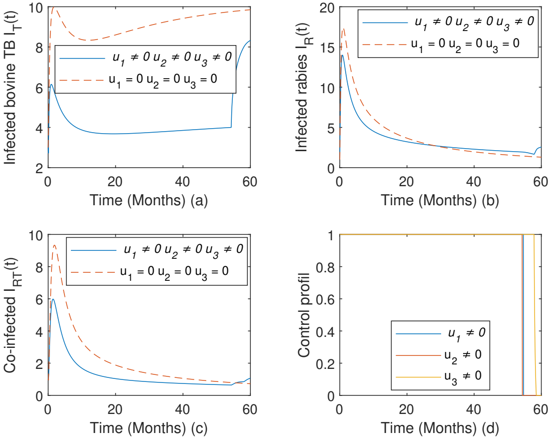

Figure 13: Effects of implementing all controls on bovine TB, rabies and their co-infection transmission dynamics, with $\alpha = 0.75$

Summary

In summary, strategy E is significantly more effective in the management of bovine TB, but performs less well in the control of rabies and co-infection. In contrast, strategy B offers satisfactory control across the three infection scenarios, although it is somewhat inadequate for bovine tuberculosis.

Conclusion

In this presentation, realistic deterministic fractional differential equations (FDEs) based on compartmental models for the transmission of bovine tuberculosis, rabies and their co-infection within a population have been developed and analyzed. These formulated models were examined as autonomous systems, and the co-infection model was extended to optimal control problems. The fundamental properties of the models showed that their solutions are positive and bounded, and there exist regions where the models are well posed both mathematically and epidemiologically. Furthermore, these models have been studied both theoretically and numerically. The autonomous co-infected model was also transformed into an optimal control problem informed by sensitivity analysis. Applying Pontryagin's maximum principle, the Hamiltonian, adjoint variables, control characterizations, and the optimality conditions were derived. To find the most effective method to eliminate the co-infection of bovine tuberculosis and rabies from the community, the optimal control problem was numerically addressed using a forward-backward predict-evaluate correct-evaluate (PECE) algorithm. The simulation results indicate that the second strategy (combining prevention efforts for bovine tuberculosis with vaccination with rabies) is the most effective intervention.

Necessary conditions for an optimal control

For recall, the Euler-Lagrange equations for the Fractional Optimal Control Problem (FOCP) are provided in Equations (41,42, 43) discused ealier. These equations state the prerequisites that must be met for the FOCP under study to function optimally. The key distinction between them and the Euler-Lagrange equations used in traditional Optimal Control Problems (OCPs) is the inclusion of both left and right fractional derivatives in the differential equations that are produced. It is important to note that although Equation (42) uses the right Caputo fractional derivative (CFD), Equation (41) uses the left CFD. Therefore, in order to account for all boundary conditions, finding a solution to the FOCP requires not only knowledge of the forward derivatives but also the backward derivatives [Kheiri and Jafari, 2018].
The necessary conditions for the FOCP Optimality are given in (1).

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